Retina Eye Disease Detection

CSET485 - Case Study Based Project |

Team No. S3-T23

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Implementation and Technical Execution

This project is based on the technical base of a multi output Convolutional Neural Network architecture. Such a design enables one model to be able to complete two different diagnostic tasks with a single input image. The model employs common layers of feature extraction, which learn to recognize complex patterns in the retina. These involve the identification of anatomical structures such as the optic disc and macula, and detection of abnormality such as lesions, swelling of vessels or clouding. Through common backbone, the system is more effective compared to running two distinct models.

Core Architectural Components

The system is changed to shared layers to two specialized heads giving categorical and continuous data.

- Shared Convolutional Layers: These layers are made up of conv2D, ReLU activation, and MaxPooling blocks. Their task is to study spatial properties beginning with simple edges up to complicated textures and disease signs.
- Classification Head: This branch involves fully connected dense layer and then a Softmax activation function. It classifies the fundus image into either of the four categories: Cataract, Diabetic Retinopathy, Glaucoma or Normal.
- Regression Head: This is another branch that is concerned with severity estimation. It gives a continuous value, which describes the level of progression of the identified condition.
- Preprocessing Pipeline: Each image uploaded is subjected to a stringent transformation. It is transformed to RGB color space and is resized to 224 by 224 pixels. The pixel values are then divided by 255 to scale them into a range of [0, 1].
- Optimization Strategy: The model is optimized with the weighted average of two loss functions. The classification task is handled with categorical Cross Entropy and the severity regression is handled with Mean Squared Error.**Mathematical Formulation**

The mathematical reasoning makes the model results mathematically and clinically useful. The head of the classification is based on the Softmax:

$$P(class_i) = \backslash frac\{e^{\{z_i\}}\}\left\{\sum_j e^{\{z_j\}}\right\}$$

This formula transforms raw network scores into probabilities which add up to one. In the estimation of the severity, the result of the regression is brought to a 0 to 1 scale by a regression clipping and scaling formula:

$$Severity \ = \ clip\left(\frac{raw_{output} + \ 1}{2}, 0, 1\right)$$

This makes the score of 0 reflect a healthy retina and the score of 1 mean maximum disease severity.

Results and Analysis

High resolution retinal fundus photographs were used to test the performance of the system. The data was split into 70 percent training, 15 percent validation and 15 percent final testing. This strict division guarantees that the metrics that were recorded are representative of the model on absolutely new clinical data.

Detailed Performance Metrics

The model had high accuracy on all the four disease categories. The system accuracy was found to be 92.9 percent overall, which is competitive in automated screening tool.

- Normal Class: This group had the best F1 of 96.3 percent. This indicates that the model is very effective in detecting healthy eyes and minimizing false alarms.
- Cataract Detection: A 94.2 percent F1 score was recorded by the system on Cataracts, which is able to detect the gradual blurring of the lens.
- Diabetic Retinopathy: The class scored a 91.5 percent F1. The model was able to do this, detecting the slightest indicators of vessel damage due to high blood sugar.
- Glaucoma: This was determined as the most difficult with F1 of 89.6 percent. The challenge lies in the fact that the loss is caused by slight variations in the optic disc which may be difficult to detect in early ages.
- Severity Regression: Mean Squared Error of regression head was 0.042. This low error has proven this model to be very precise in estimating the stage of a disease.

Analytical Observations

The training process was analysed to reveal that multi task learning was a valuable advantage. Simultaneous training on classification and severity did not impair the execution of either of the tasks. Rather, the shared feature learning seemed to make the model generalize to a better degree than different images. It is reported that the model is not over predicting or missing many cases as the balanced precision of 92.6 percent and the recall of 93.3 percent show that the model is not over predicting or missing a large number of cases.

Demonstration and Explanation

The project is represented by an interactive web application created with the help of the Streamlit framework. Such deployment option exposes the sophisticated AI model to healthcare professionals via any internet web browser. The interface will be user-friendly and will not need any technical knowledge of the user.

System Workflow and User Experience

The application workflow is predicated upon pragmatism and clarity of information in clinical settings.

- Image Upload: By clicking on a browse button or dragging a JPEG or PNG fundus image into the upload widget.
- Instant Preprocessing: It instantly resizes the image and normalizes the pixel data to values with no manual input from the technician.
- Inference Cycle: The image goes through the shared CNN layers. Also, the two heads provide classification probabilities and severity scores. From start to finish, the entire cycle takes less than two seconds to complete.
- Visual Results: The screen shows its diagnosis as well as the percentage of confidence in this outcome.
- Severity Visualization: A progress bar indicates severity on a scale from 0 to 1. This effectively conveys the severity of the case to the doctor at the first glance.
- Probability Chart: Transparency in the model's decision making is achieved by displaying a bar chart showing the probability for all four classes. **Clinical Advice Integration**

The demonstration also highlights how the system translates data into medical guidance. Based on the specific combination of the detected disease and its severity score, the app surfaces tailored recommendations. For example, if the model detects Diabetic Retinopathy with a severity score above 0.7, it automatically suggests an immediate specialist referral. This ensures that the results are not just numbers but actionable insights for the clinician.

Societal Benefits and Healthcare Equity

The impact of this technology is worth more than its technical feat to its social value.

- Catch it Early Vision loss is final: Conditions like Glaucoma have no ache and are not noticed till peripheral vision is lost. With this system, early screening is made easy and accessible, thus preventing irreversible blindness.
- Reducing Specialist Burden: AI pre screening serves as a smart filter. Instead of seeing thousands of routine, healthy images, specialists can concentrate their limited time on urgent cases with high risks.

- Access to Underserved Areas: Since the tool is browser based, it can be accessed in rural clinics where specialists are rarely available. The health workers in this area can carry out screening processes with little training.
- Cost Effective Screening: The use of lightweight deployment tools such as Streamlit does away with costly software installations or expensive local servers.

Future Scope and Project Evolution

The current system can be used as a basis for multiple future improvements, which will make the system more useful for clinical purposes.

- Grad-CAM Explainability: Later versions could add heatmaps that can be superimposed on the fundus image to highlight what regions contributed to the outcome.
- Attention Mechanisms: Self attention layers could be added to the model to enable it to dynamically attend to the most relevant parts of the retina for each disease.
- Mobile Deployment: Retraining in mobile apps might make it possible for community health workers to conduct screenings offline in remote areas without internet access.
- Electronic Health Record Integration: The system's connection to electronic patient records would enable automatic recording of severity trends and diagnostic history.

This groundwork in multi output learning and live deployment provides a scalable platform to better global eye care and prevent millions of cases of avoidable vision loss.

Conclusion and Societal Impact Discussion

This project delivers a comprehensive system for retinal health monitoring. Combining diagnosis and severity estimation in a single browser based tool has covered a crucial gap of modern ophthalmic AI. The system proves that it is possible to achieve medical grade performance without the need for expensive or specialized local hardware.